

Label-permutation negative control

Observed $\rho = 0.678$

Empirical $p < 0.0002$

What is permuted?

True labels

$\rho = 0.68$

Shuffled labels

$\rho \approx 0$

Sequences fixed; labels shuffled
(not sequence permutation)

Density



Spearman ρ (sequence-level vs Spillover Total)

Regression model; overlap cohort $n = 632$. Null reconstructed from reported mean/sd.